Hongjin Chen

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Education

McGill University — Montreal, Canada
PhD in Animal Science: Dairy Milk Analysis; Milk Composition; Dairy Production — Jan. 2026 – Exp. Jan. 2029
Supervisor: Dr. Matthias Klein

Technical University of Denmark — Copenhagen, Denmark
MSc Eng in Bioinformatics and System Biology — Aug. 2023 – Oct. 2025
GPA: 10.1/12 (3.7/4.0)
Main courses: Machine Learning; Deep Learning; High Performance Computing; Next-Generation Sequencing

McGill University — Montreal, Canada
Exchange in Faulty of Science — Jan. 2025 – Apr. 2025

Sichuan Agricultural University — Chengdu, China
BSc in Veterinary Medicine — Sept. 2018 – June 2022
GPA: 3.7/4.0; Class Rank: 1/35; Major Rank: 17/262

Publication & Conference

- Nourbakhsh, M., Zheng, Y., Noor, H., **Chen, H.**, Akhuli, S., Tiberti, M., Gevaert, O. and Papaleo, E., 2025. Revealing cancer driver genes through integrative transcriptomic and epigenomic analyses with Moonlight. PLOS Computational Biology, 21(4), p.e1012999.
- Leuchte K, Luu TV, Fresnillo Saló S, Madsen K, Heide-Ottosen L, Skadborg SK, Kemming JS, Holmström MO, **Chen H**, Olsen LR, Vinther A. Moving for optimal immunity: the effect of acute high-intensity interval training on phenotype, virus specificity and chemokine receptor expression in human CD8+ T cells. Frontiers in Immunology. 2025;16:1739657.
- Leuchte K, Madsen K, **Chen H**, Olsen LR, Olofsson GH. Acute high-intensity interval training induces remodeling of the immune-specific serum proteome with age-dependent modulation. Proteomics. Under review.
- **Chen H**, Christian LM, Greenwood P, Papaleo E, Klein MS. Serum glucose depletion and reduced ketogenesis in postpartum weight retention revealed by machine learning metabolomics analysis. BBA - Molecular Basis of Disease. Submitted.
- **Chen, H.**, Christian, L. M., Greenwood, P., Papaleo, E., Klein, M. S. (2026, April 30–May 1). Metabolomic serum biomarkers and machine learning improve prediction of postpartum weight retention [**Oral and poster presentation**]. 7th Annual Canadian Metabolomics Conference, Toronto, Canada.
- **Chen, H.**, Akhuli, S., Nourbakhsh, M., Tiberti, M., & Papaleo, E. (2024, August 21–22). Prediction of driver genes in basal-like breast cancer with Moonlight2R [Poster presentation]. 8th Annual Danish Bioinformatics Conference, Copenhagen, Denmark.

Research Experience

Graduate Research Trainee — May 2025 - Sept. 2025
Supervisor: Dr. Matthias Klein, McGill University — Montreal, Canada

- **Machine Learning Modeling & NMR Data Analysis:** Conducted end-to-end machine learning modeling for metabolomic biomarker prediction in postpartum weight retention (PWR) using NMR data, using nine algorithms including Logistic Regression, Gaussian and Bernoulli Naive Bayes, SVM, Random Forest, KNN, XGBoost, Decision Tree, and MLP, with extensive cross-validation and hyperparameter tuning.
- **Evaluation Model Performance & Biological Analysis:** Developed two-layer nested cross validations framework and biological pathway interpretation (KEGG enrichment via MetaboAnalyst) to assess model performance and interpret feature contributions.

Academic Researcher — Oct. 2024 - Mar. 2025
Supervisor: Dr. Mathias Middelboe. University of Copenhagen — Helsingør, Denmark

- **High-Performance Computing & Genomic Analysis:** Large-scale WGS data processing on HPC infrastructure, developing end-to-end workflows (QC/trimming with FastQC/Trimmomatic, de novo assembly via SPAdes, reference-guided scaffolding with RagTag, functional annotation using Prokka/Pharokka), and characterization of plasmids (PlasmidFinder) and virulence/resistance genes (ABRicate/ResFinder) through integrated pipelines, while optimizing resource allocation through conda environments and qsub job scheduling.
- **Comparative Genomics & Evolutional Dynamics:** Conducted pangenome (Roary) and SNP-based phylogenetic analyses (BWA-MEM2/BCFtools/RAxML) to resolve bacteria/phage evolutionary dynamics, with structural variant detection (Mauve/Mafft) analysis.

Student Researcher Sept. 2024 – Jan. 2025

Supervisor: Dr. Lars Rønn Olsen. Technical University of Denmark & University of Copenhagen Copenhagen, Denmark

- **Immunological Data Analysis & Modeling:** Conducted data analysis on immunological datasets using R, applying statistical and machine learning techniques including LASSO regression, Random Forest, and Neural Networks to model and predict CD8 mobilization and protein expression levels in immunological responses in post-exercise.
- **Protein Expression Profiling:** Performed differential expression and enrichment analyses using Limma and Gene Set Enrichment Analysis (GSEA) to identify key protein pathways and variables influencing immune responses.

Student Researcher Mar. 2024 – Sept. 2024

Supervisor: Dr. Elena Papaleo. Technical University of Denmark & Danish Cancer Institute Copenhagen, Denmark

- **Driver Gene Prediction in Basal-like Breast Cancer:** Conducted differential expression analysis to identify differentially expressed genes (DEGs) in basal-like breast cancer. Integrated different mutation data, and mechanistic indicators in Moonlight2R for driver gene discovery in breast cancer, providing insights into key genetic drivers of cancer progression.
- **Programming in R for Bioconductor Packages:** Developed and integrated new analysis scripts into the Moonlight2R pipeline.

Teaching Experience

Teaching Assistant in ANSC 514 Sept. 2026 - Dec. 2026

Course: Coding in Production Data Montreal, Canada

- Assisted students with coding in R, including R programming and introductory machine learning applications.
- Provided course support by answering questions, troubleshooting code, and helping students understand programming concepts and analytical workflows.

Awards

- International Fee Exemptions ($50,000) - Quebec Ministry of Education Sept. 2026
- GREAT Award ($900) - McGill University Mar. 2026
- Amy Wong Scholarship ($48,500) - McGill University Jan. 2026
- Graduate Excellence Fellowships ($27,200) - McGill University Jan. 2026
- International Student Ambassador - Technical University of Denmark Feb. 2025
- Licensed Veterinarian - Ministry of Agricultural and Rural affairs of P.R.China Nov. 2023
- Outstanding Undergraduate Student - Sichuan Agricultural University June 2022
- Outstanding Student - Sichuan Agricultural University 2021 & 2022
- Ningbo Sansheng Student Scholarship - Sichuan Agricultural University Nov. 2021
- Outstanding Young Volunteers - Sichuan Agricultural University May 2021
- Guilin Liyuan Student Scholarship - Sichuan Agricultural University Nov. 2020
- Outstanding Student Leader - Sichuan Agricultural University Nov. 2020

Skills

- Programming: R (RStudio), Python (Numpy, Pandas, Scikit-learn, PyTorch, Matplotlib.), Linux shell scripting (HPC based), SQL.
- Data Analysis: Machine Learning (LR, DT, RF, SVM, KNN, GBM), Deep learning (ANN,CNNs, RNNs), Data Visualization.
- Language: Fluent in both Chinese and English, French (Beginner).
- Interests: Soccer referee, traveling, climbing.